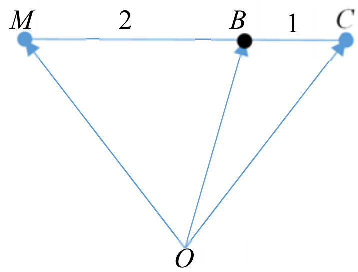
Qn	Solution
1	Let e, g, w be amount of electricity and gas in kWh and water in m^3 used by Mr Lim in June respectively. $0.1775e+0.1715g+2.852w=143.06-36$ $0.1775e+0.1715g+2.852w=107.06 \quad ----(1)$ $0.2365e+0.2279g+3.672w=140.78 \quad ----(2)$ $0.1762e+0.1698g+2.741w=144.96-40$ $0.1762e+0.1698g+2.741w=104.96 \quad ----(3)$ By G.C., $e=266.4558551=266$ $g=115.0629925=115$ $w=14.03603875=14.0$ Mr Lim used 266kWh of electricity, 115kWh of gas and 14.0 m^3 of water in June.
2(i)	$\frac{\mathrm{d}}{\mathrm{d}x}\tan^{-1}\left(x^k\right)=\frac{kx^{k-1}}{1+x^{2k}}$
2(ii)	$u=\tan^{-1}\left(x^k\right) \qquad v'=x^{k-1}$ $u'=\frac{kx^{k-1}}{1+x^{2k}} \qquad v=\frac{x^k}{k}$ $\int x^{k-1}\tan^{-1}\left(x^k\right)\mathrm{d}x$ $=\frac{x^k}{k}\tan^{-1}\left(x^k\right)-\int\frac{x^{2k-1}}{1+x^{2k}}\mathrm{d}x$ $=\frac{x^k}{k}\tan^{-1}\left(x^k\right)-\frac{1}{2k}\int\frac{2kx^{2k-1}}{1+x^{2k}}\mathrm{d}x$ $=\frac{x^k}{k}\tan^{-1}\left(x^k\right)-\frac{1}{2k}\ln\left(1+x^{2k}\right)+c$ since $1+x^{2k}>0$
3(i)	$k=\frac{ax+b}{x^2+1}$ $kx^2+k=ax+b$ $kx^2-ax+(k-b)=0 \quad$ ------------(1) (shown)
3(ii)	For (1) to have real roots, $(-a)^2-4k(k-b)\geq 0$ $a^2-4k^2+4kb\geq 0$ $4k^2-4bk-a^2\leq 0$ Find critical values

Qn	Solution
	$k=\dfrac{4b\pm\sqrt{16b^2-4(4)(-a^2)}}{2(4)}$ $k=\dfrac{b\pm\sqrt{b^2+a^2}}{2}$ Therefore $4k^2-4bk-a^2\le 0$ $\dfrac{b-\sqrt{b^2+a^2}}{2}\le k\le\dfrac{b+\sqrt{b^2+a^2}}{2}$ Since $-1\le k\le 4$, $\dfrac{b}{2}=\dfrac{-1+4}{2}\Rightarrow b=3$ $\dfrac{b+\sqrt{b^2+a^2}}{2}=4\Rightarrow\sqrt{9+a^2}=5\Rightarrow a=\pm 4$
4	Let $a_1=b_1=a$ $a_3=b_3 \Rightarrow a+2d=a\,r^2$ ------------(1) $a_7=b_5 \Rightarrow a+6d=a\,r^4$ ------------(2) Eliminate d, (1)×3 – (2): $2a=3a\,r^2-a\,r^4$ $a\left(r^4-3\,r^2+2\right)=0$ Since $a\ne 0$, $r^4-3r^2+2=0$ -------------(3) $\Rightarrow r^2=1,\quad\text{or}\quad r^2=2$ Since $r>0,\ r\ne 1$, therefore, $r=\sqrt{2}$. Sub $r=\sqrt{2}$ into (1): $a+2d=2a \Rightarrow d=\dfrac{a}{2}$ If $a_n=b_m$:

Qn	Solution
	$a+(n-1)d = a\,r^{m-1}$ $a+(n-1)\dfrac{a}{2} = a\,r^{m-1}$ $1+\dfrac{(n-1)}{2} = r^{m-1}$ $\dfrac{(n+1)}{2} = 2^{\frac{1}{2}(m-1)}$ $n = 2^{\frac{1}{2}(m+1)} - 1$
5(a)	Area of $ABC = \dfrac{1}{2}\left\lvert \overrightarrow{AC}\times\overrightarrow{BC}\right\rvert$ $= \dfrac{1}{2}\left\lvert\left(-\sqrt{2}\mathbf{a}-\mathbf{b}-\mathbf{a}\right)\times\left(-\sqrt{2}\mathbf{a}-\mathbf{b}-\mathbf{b}\right)\right\rvert$ $= \dfrac{1}{2}\left\lvert\left(\left(-\sqrt{2}-1\right)\mathbf{a}-\mathbf{b}\right)\times\left(-\sqrt{2}\mathbf{a}-2\mathbf{b}\right)\right\rvert$ $= \dfrac{1}{2}\left\lvert\left(2+\sqrt{2}\right)\ \mathbf{a}\times\mathbf{a}+\sqrt{2}\ \mathbf{b}\times\mathbf{a}+\left(2\sqrt{2}+2\right)\ \mathbf{a}\times\mathbf{b}+2\ \mathbf{b}\times\mathbf{b}\right\rvert$ $= \dfrac{1}{2}\left\lvert\sqrt{2}\ \mathbf{b}\times\mathbf{a}+\left(2\sqrt{2}+2\right)\ \mathbf{a}\times\mathbf{b}\right\rvert$ $= \dfrac{1}{2}\left\lvert\left(\sqrt{2}+2\right)\ \mathbf{a}\times\mathbf{b}\right\rvert$ $= \dfrac{\sqrt{2}+2}{2}\lvert\mathbf{a}\times\mathbf{b}\rvert$ $\therefore k = \dfrac{\sqrt{2}+2}{2}$ Since $\angle AOB = 2\times\angle ACB$ since $\angle$ at centre = $2\times\angle$ at circumference. Area of $ABC = \dfrac{2+\sqrt{2}}{2}\lvert\mathbf{a}\times\mathbf{b}\rvert$ $= \dfrac{2+\sqrt{2}}{2}\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\left\lvert\sin\dfrac{3\pi}{4}\right\rvert\lvert\hat{\mathbf{n}}\rvert$ $= \dfrac{2+\sqrt{2}}{2}(1)(1)\left(\dfrac{\sqrt{2}}{2}\right)(1)$ $= \dfrac{1+\sqrt{2}}{2}$ units^2

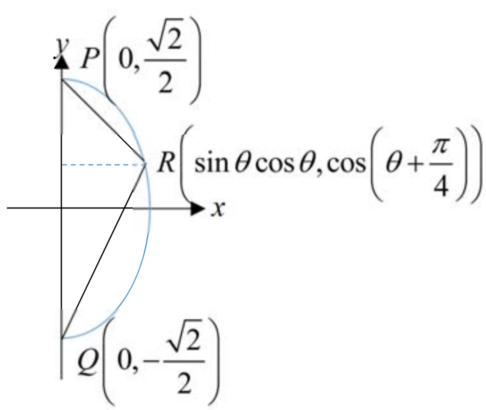
Qn	Solution
5(b)	Since $\overrightarrow{MC}=3\overrightarrow{BC}$, $\overrightarrow{OB}=\dfrac{\overrightarrow{OM}+2\overrightarrow{OC}}{3}$ $\overrightarrow{OM}=3\overrightarrow{OB}-2\overrightarrow{OC}$ $=3\mathbf{b}-2\left(-\sqrt{2}\mathbf{a}-\mathbf{b}\right)$ $=2\sqrt{2}\mathbf{a}+5\mathbf{b}$ 
6(i)	$\dfrac{\mathrm{d}}{\mathrm{d}\theta}\sec^n\theta=n\sec^{n-1}\theta\left(\sec\theta\tan\theta\right)$ $=n\sec^n\theta\tan\theta$
6(ii)	$x=2\tan^2\theta \Rightarrow \dfrac{\mathrm{d}x}{\mathrm{d}\theta}=4\tan\theta\sec^2\theta$ When $x=0$, $\tan\theta=0 \Rightarrow \theta=0$ When $x=2$, $\tan\theta=1 \Rightarrow \theta=\dfrac{\pi}{4}$ $\int_0^2 x\sqrt{1+\dfrac{x}{2}}\,\mathrm{d}x$ $=\int_0^{\frac{\pi}{4}}2\tan^2\theta\sqrt{1+\tan^2\theta}\left(4\tan\theta\sec^2\theta\right)\mathrm{d}\theta$ $=\int_0^{\frac{\pi}{4}}8\tan^3\theta\left\|\sec\theta\right\|\sec^2\theta\ \mathrm{d}\theta$ $=\int_0^{\frac{\pi}{4}}8\tan^3\theta\sec^3\theta\ \mathrm{d}\theta\ \left(\because \sec\theta>0 \text{ for } 0<\theta<\dfrac{\pi}{4}\right)$ $=8\int_0^{\frac{\pi}{4}}\tan\theta\left(\sec^2\theta-1\right)\sec^3\theta\ \mathrm{d}\theta$ $=8\int_0^{\frac{\pi}{4}}\sec^5\theta\tan\theta-\sec^3\theta\tan\theta\ \mathrm{d}\theta$ where $k=8$, $a=0$, $b=\dfrac{\pi}{4}$

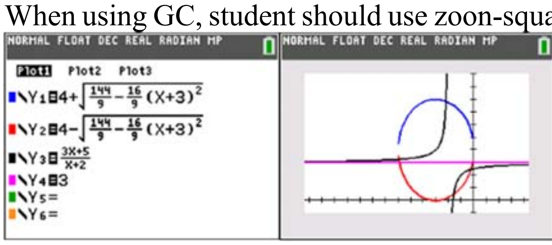
Qn	Solution
6(iii)	$\int_0^2 x\sqrt{1+\frac{x}{2}}\ \mathrm{d}x$ $=8\int_0^{\frac{\pi}{4}} \sec^5\theta\tan\theta-\sec^3\theta\tan\theta\ \mathrm{d}\theta$ $=8\left[\frac{\sec^5\theta}{5}-\frac{\sec^3\theta}{3}\right]_0^{\frac{\pi}{4}}$ $=8\left(\frac{\left(\sqrt{2}\right)^5}{5}-\frac{\left(\sqrt{2}\right)^3}{3}\right)-8\left(\frac{1}{5}-\frac{1}{3}\right)$ $=\frac{16}{15}\left(\sqrt{2}+1\right)$ $A=\frac{16}{15}, B=1$
7(i)	To find exact points of intersection, sub C_2 into C_1: $x^2+\frac{4}{1+x^2}=4$ $x^2+x^4+4=4+4x^2$ $x^4-3x^2=0$ $x^2\left(x^2-3\right)=0$ $x=0$ or $x=\pm\sqrt{3}$ (or using G.C,) When $x=0$, $y^2=4\Rightarrow y=\pm2$ When $x=\pm\sqrt{3}$, $y^2=1\Rightarrow y=\pm1$ $\therefore (0,2),(0,-2),\left(\sqrt{3},1\right),\left(-\sqrt{3},1\right),\left(\sqrt{3},-1\right),\left(-\sqrt{3},-1\right)$
7(ii)	<u>**Method 1**</u>: Using $\int x\ \mathrm{d}y$ Area of shaded region $=4\left[\int_0^1\sqrt{4-y^2}\ \mathrm{d}y+\int_1^2\sqrt{\frac{4}{y^2}-1}\ \mathrm{d}y\right]$ $=11.260\text{ units}^2$ (to 3d.p) OR $=4\pi-4\int_1^2\sqrt{4-y^2}-\sqrt{\frac{4}{y^2}-1}\ \mathrm{d}y$ <u>**Method 2**</u>: Using $\int y\ \mathrm{d}x$

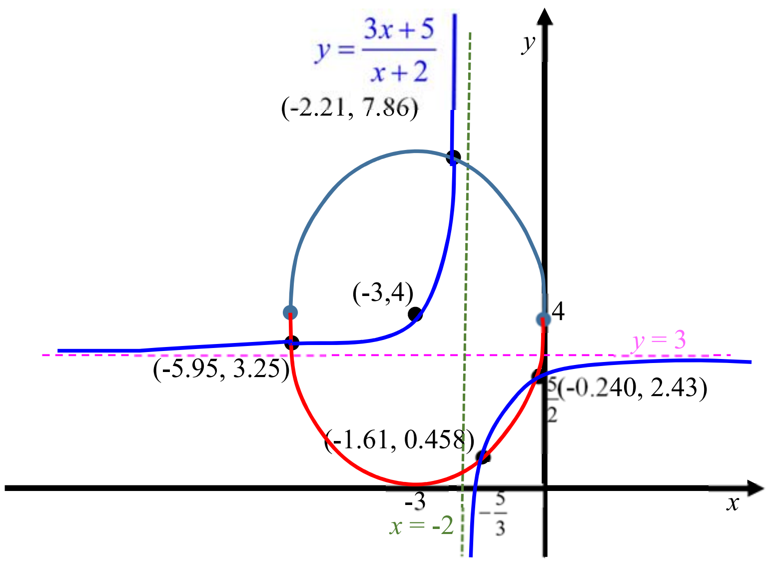
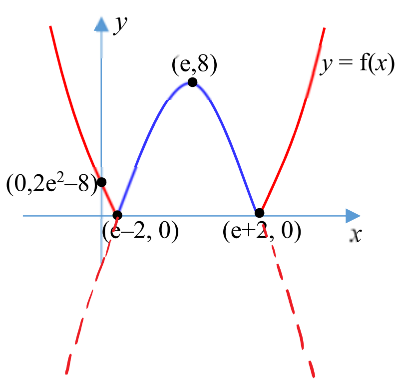
Qn	Solution
	Area of shaded region $= 4\left[\int_0^{\sqrt{3}} \frac{2}{\sqrt{1+x^2}}\,\mathrm{d}x + \int_{\sqrt{3}}^{2} \sqrt{4-x^2}\,\mathrm{d}x\right]$ $= 11.260 \text{ units}^2 \left(\text{to 3d.p}\right)$ OR $= 4\pi - 2\int_{-\sqrt{3}}^{\sqrt{3}} \sqrt{4-x^2} - \sqrt{\frac{4}{1+x^2}}\,\mathrm{d}x$
7(iii)	$V_x = 2\pi\left[\int_0^{\sqrt{3}} \frac{4}{1+x^2}\,\mathrm{d}x + \int_{\sqrt{3}}^{2} 4-x^2\,\mathrm{d}x\right]$ $= 2\pi\left[\left[4\tan^{-1}x\right]_0^{\sqrt{3}} + \left[4x - \frac{x^2}{3}\right]_{\sqrt{3}}^{2}\right]$ $= 2\pi\left(\frac{4\pi}{3} + 8 - \frac{8}{3} - 4\sqrt{3} + \sqrt{3}\right)$ $= \frac{2\pi}{3}\left(4\pi + 16 - 9\sqrt{3}\right) \text{ units}^3$ OR $V_x = \frac{4}{3}\pi(2)^3 - \pi\left[\int_{-\sqrt{3}}^{\sqrt{3}} 4 - x^2 - \frac{4}{1+x^2}\,\mathrm{d}x\right]$
8(i)	From RHS: $\frac{1}{2}\left(\frac{1}{\sqrt{2r+1}} - \frac{1}{\sqrt{2r+3}}\right) = \frac{1}{2}\frac{\sqrt{2r+3}-\sqrt{2r+1}}{\sqrt{2r+1}\sqrt{2r+3}}$ $= \frac{1}{2}\frac{\left(\sqrt{2r+3}-\sqrt{2r+1}\right)\left(\sqrt{2r+3}+\sqrt{2r+1}\right)}{\sqrt{2r+1}\sqrt{2r+3}\left(\sqrt{2r+3}+\sqrt{2r+1}\right)}$ $= \frac{1}{2}\frac{(2r+3)-(2r+1)}{(2r+3)\sqrt{(2r+1)} + (2r+1)\sqrt{(2r+3)}}$ $= \frac{1}{(2r+3)\sqrt{(2r+1)} + (2r+1)\sqrt{(2r+3)}}$ =LHS OR From LHS: $\frac{1}{(2r+3)\sqrt{(2r+1)} + (2r+1)\sqrt{(2r+3)}}$ $= \frac{1}{\sqrt{(2r+3)}\sqrt{(2r+1)}\left[\sqrt{(2r+3)}+\sqrt{(2r+1)}\right]}$

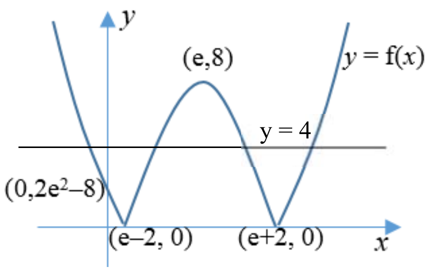
Qn	Solution
	$=\dfrac{\sqrt{(2r+3)}-\sqrt{(2r+1)}}{\sqrt{(2r+3)}\sqrt{(2r+1)}\ \left[2r+3-2r-1\ \right]}$ $=\dfrac{\sqrt{2r+3}-\sqrt{2r+1}}{2\sqrt{2r+1}\sqrt{2r+3}}$ $=\dfrac{1}{2}\left(\dfrac{1}{\sqrt{2r+1}}-\dfrac{1}{\sqrt{2r+3}}\right)$ =RHS
8(ii)	$S_n=\displaystyle\sum_{r=1}^{n}\frac{1}{(2r+3)\sqrt{(2r+1)}+(2r+1)\sqrt{(2r+3)}}$ $=\dfrac{1}{2}\displaystyle\sum_{r=1}^{n}\left(\frac{1}{\sqrt{(2r+1)}}-\frac{1}{\sqrt{(2r+3)}}\right)$ ---------------(1) $=\dfrac{1}{2}\left[\begin{array}{l}\dfrac{1}{\sqrt{3}}-\cancel{\dfrac{1}{\sqrt{5}}}\\ \cancel{\dfrac{1}{\sqrt{5}}}-\cancel{\dfrac{1}{\sqrt{7}}}\\ \ldots\\ \cancel{\dfrac{1}{\sqrt{2n-1}}}-\cancel{\dfrac{1}{\sqrt{2n+1}}}\\ \cancel{\dfrac{1}{\sqrt{2n+1}}}-\dfrac{1}{\sqrt{2n+3}}\end{array}\right]$ $=\dfrac{1}{2}\left(\dfrac{1}{\sqrt{3}}-\dfrac{1}{\sqrt{2n+3}}\right)$
8(iii)	$n\to\infty,\quad \dfrac{1}{\sqrt{2n+3}}\to 0$ $\displaystyle\lim_{x\to\infty}S_n=\frac{1}{2\sqrt{3}}$ $\left\|S_n-\dfrac{1}{2\sqrt{3}}\right\|<0.05$ $\left\|\dfrac{1}{2}\left(\dfrac{1}{\sqrt{3}}-\dfrac{1}{\sqrt{2n+3}}\right)-\dfrac{1}{2\sqrt{3}}\right\|<0.05$ ------------(2) $\left\|\dfrac{1}{2\sqrt{2n+3}}\right\|<0.05$

Qn	Solution
	$\frac{1}{2\sqrt{2n+3}}<0.05$ $\sqrt{2n+3}>10$ $2n+3>100$ $n>48.5$ OR From GC, $n=48$, $\frac{1}{2\sqrt{2n+3}}=0.05025$ $n=49$, $\frac{1}{2\sqrt{2n+3}}=0.04975$ therefore the least value n is 49.
9(i)	$\frac{\mathrm{d}x}{\mathrm{d}t}=\cos^2 t-\sin^2 t$ $\frac{\mathrm{d}y}{\mathrm{d}t}=\frac{\mathrm{d}}{\mathrm{d}t}\left(\cos t\cos\frac{\pi}{4}-\sin t\sin\frac{\pi}{4}\right)$ $=\frac{\sqrt{2}}{2}\frac{\mathrm{d}}{\mathrm{d}t}(\cos t-\sin t)$ $=-\frac{\sqrt{2}}{2}(\sin t+\cos t)$ $\therefore\frac{\mathrm{d}y}{\mathrm{d}x}=\frac{\mathrm{d}y}{\mathrm{d}t}.\frac{\mathrm{d}t}{\mathrm{d}x}$ $=\frac{-\frac{\sqrt{2}}{2}(\sin t+\cos t)}{\cos^2 t-\sin^2 t}$ $=-\frac{\sin t+\cos t}{\sqrt{2}(\cos t+\sin t)(\cos t-\sin t)}$ $=\frac{1}{\sqrt{2}(\sin t-\cos t)}$ (shown)
9(ii)	For tangent parallel to $y-$axis, $\frac{\mathrm{d}y}{\mathrm{d}x}$ is undefined $\Rightarrow$ $\sin t-\cos t=0$ $\tan t=1$ $t=\frac{\pi}{4}$

Qn	Solution
	$\Rightarrow$ Equation of tangent is: $x=\sin\frac{\pi}{4}\cos\frac{\pi}{4}=\left(\frac{\sqrt{2}}{2}\right)^2=\frac{1}{2}$ $\therefore x=\frac{1}{2}$
9(iii)	Method 1 When $x=0$, $\sin t\cos t=0$ $2\sin t\cos t=0$ $\sin 2t=0$ $2t=0 \ \ or \ \ \pi$ $t=0\Rightarrow y=\cos\left(0+\frac{\pi}{4}\right)=\frac{\sqrt{2}}{2}$ $or\ t=\frac{\pi}{2}\Rightarrow y=\cos\left(\frac{\pi}{2}+\frac{\pi}{4}\right)=-\frac{\sqrt{2}}{2}$ Method 2 When $x=0$, $\sin t\cos t=0$ $\sin t=0\Rightarrow t=0\Rightarrow y=\cos\left(0+\frac{\pi}{4}\right)=\frac{\sqrt{2}}{2}$ $\cos t=0\Rightarrow t=\frac{\pi}{2}\Rightarrow y=\cos\left(\frac{\pi}{2}+\frac{\pi}{4}\right)=-\frac{\sqrt{2}}{2}$ $\therefore P\left(0,\frac{\sqrt{2}}{2}\right), Q\left(0,-\frac{\sqrt{2}}{2}\right)$
9(iv)	 From the diagram, area of triangle PQR is given by

Qn	Solution
	$\frac{1}{2}\times 2\left(\frac{\sqrt{2}}{2}\right)\times\sin\theta\cos\theta$ $=\frac{\sqrt{2}}{2}\left(\frac{1}{2}\sin 2\theta\right)$ $=\frac{\sin 2\theta}{2\sqrt{2}}$ $=\frac{\sqrt{2}\sin 2\theta}{4}\text{ (shown)}$ Since $0\le\sin 2\theta\le 1$ for $0\le\theta\le\frac{\pi}{2}$, max area of triangle PQR is when $\sin 2\theta=1\Rightarrow\theta=\frac{\pi}{4}$
10 (i)	$y=\frac{3x+5}{x+2}=3-\frac{1}{x+2}$ <u>Method 1:</u> $\frac{1}{x}\xrightarrow{1}\frac{1}{x+2}\xrightarrow{2}-\frac{1}{x+2}\xrightarrow{3}3-\frac{1}{x+2}$ 1. Translate 2 units in the negative x-direction 2. Reflect in x-axis 3. Translate 3 units in the positive y-direction <u>Method 2:</u> $\frac{1}{x}\rightarrow-\frac{1}{x}\rightarrow-\frac{1}{x+2}\rightarrow 3-\frac{1}{x+2}$ 1. Reflect in y-axis 2. Translate 2 units in the negative x-direction 3. Translate 3 units in the positive y-direction Students should recognize the equation of the ellipse. $16(x+3)^2+9(y-4)^2=144$ $\frac{(x+3)^2}{3^2}+\frac{(y-4)^2}{4^2}=1$ When using GC, student should use zoon-square to ensure that the axis are both equal in scale. 

Qn	Solution
10 (ii)	
10 (iii)	From the graph, compare the curve with the lower half of the ellipse, $\{x \in \mathbb{R} : -5.95 < x < -2 \text{ or } -1.61 < x < -0.240\}$
11 (i)	 **To find exact *x*-coordinates of *x* intercepts:** $4-(x-\mathrm{e})^2=0$ $(x-\mathrm{e})^2=4$ $x-\mathrm{e}=\pm 2$ $x=\mathrm{e}\pm 2$ **To find exact *y*-coordinates of *y* intercept:** $2\left[-4+(0-\mathrm{e})^2\right]=y$ $y=2\mathrm{e}^2-8$

Qn	Solution
	NOTE: $2\left\|4-(x-\mathrm{e})^2\right\| = \begin{cases} 2\left[4-(x-\mathrm{e})^2\right] & ,\mathrm{e}-2 \le x \le \mathrm{e}+2 \\ 2\left[-4+(x-\mathrm{e})^2\right] & ,x < \mathrm{e}-2 \text{ or } x > \mathrm{e}+2 \end{cases}$
11 (ii)	Since $\mathrm{f}(\mathrm{e}-2)=\mathrm{f}(\mathrm{e}+2)$ Therefore f is **not a one to one function**. Hence f^{-1} does not exist.  **Alternatively,** Since the **horizontal line** $y=4$ (give a specific counter-example) intersects $y=\mathrm{f}(x)$ **more than once**, therefore f is not a one to one function. Hence f^{-1} does not exist.
11 (iii)	The function f has an inverse if its domain is restricted to $a \le x < \mathrm{e}$. Smallest $a=\mathrm{e}-2$ Therefore $y=-2(x-\mathrm{e})^2+8$ $-\frac{y}{2}=(x-\mathrm{e})^2-4$ $(x-\mathrm{e})^2=4-\frac{y}{2}$ $x-\mathrm{e}=\pm\sqrt{4-\frac{y}{2}}$ $x=\mathrm{e}\pm\sqrt{4-\frac{y}{2}}$ Since $\mathrm{e}-2 \le x < \mathrm{e}$ $x=\mathrm{e}-\sqrt{4-\frac{y}{2}}$ $\mathrm{f}^{-1}: x \mapsto \mathrm{e}-\sqrt{4-\frac{x}{2}},\quad 0 \le x < 8$
11 (iv)	From part **(iii)**, $\mathrm{e}-2 \le x < \mathrm{e}$ $\mathrm{R}_{\mathrm{f}^{-1}}=\mathrm{D}_\mathrm{f}=[\mathrm{e}-2,\mathrm{e})$ $\mathrm{D}_\mathrm{g}=\mathbb{R}\setminus\{-1\}$

<table>
<tr><th>Qn</th><th>Solution</th></tr>
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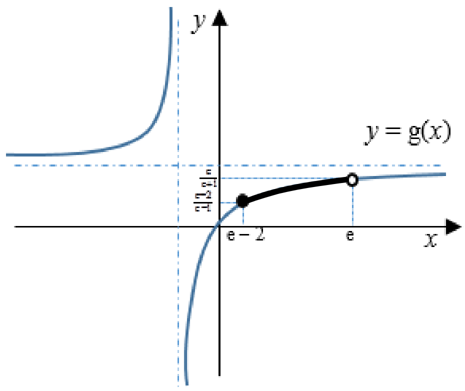
Since $\mathrm{R}_{\mathrm{f}^{-1}} \subseteq \mathrm{D}_{\mathrm{g}}$, gf^{-1} exists



$$\mathrm{R}_{\mathrm{gf}^{-1}} = \left[\frac{\mathrm{e}-2}{\mathrm{e}-1}, \frac{\mathrm{e}}{\mathrm{e}+1}\right)$$

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<tr><td>12
(i)</td><td>

Let the angle α and β be such that $\tan\alpha = \dfrac{\frac{y}{2} - 4.5 + 1.5}{x} = \dfrac{y-6}{2x}$ and

$$\tan\beta = \frac{\frac{y}{2} - 4.5 - 1.5}{x} = \frac{y-12}{2x}$$

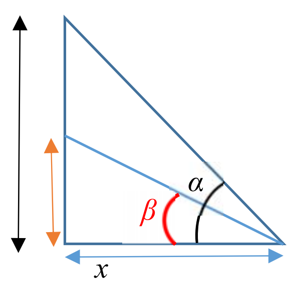
$$\tan\theta = \tan(\alpha - \beta)$$

$$= \frac{\tan\alpha - \tan\beta}{1 + \tan\alpha\tan\beta}$$

$$= \frac{\frac{y-6}{2x} - \frac{y-12}{2x}}{1 + \left(\frac{y-6}{2x}\right)\left(\frac{y-12}{2x}\right)}$$

$$= \frac{12x}{4x^2 + (y-6)(y-12)}$$

$$= \frac{12x}{4x^2 + A}$$



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<tr><td>12
(ii)</td><td>

Differentiate θ implicitly throughout with respect to x

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Qn	Solution

$$\sec^2\theta\left(\frac{\mathrm{d}\theta}{\mathrm{d}x}\right)=\frac{12\left(4x^2+A\right)-12x\left(8x\right)}{\left[4x^2+A\right]^2}$$

$$\sec^2\theta\left(\frac{\mathrm{d}\theta}{\mathrm{d}x}\right)=\frac{12\left[A-4x^2\right]}{\left[4x^2+A\right]^2}$$

$$\sec^2\theta\left(\frac{\mathrm{d}\theta}{\mathrm{d}x}\right)=\frac{12\left(\sqrt{A}-2x\right)\left(\sqrt{A}+2x\right)}{\left[4x^2+A\right]^2}$$

For $\frac{\mathrm{d}\theta}{\mathrm{d}x}=0$,

$$x=\frac{\sqrt{A}}{2}\quad(x>0)$$

Using 1st Derivative Test below,

x	$\frac{\sqrt{A}}{2}^{-}$	$\frac{\sqrt{A}}{2}$	$\frac{\sqrt{A}}{2}^{+}$
$\frac{\mathrm{d}\theta}{\mathrm{d}x}$	+ve	0	-ve
Shape of tangent	/	—	\

Thus for maximum θ, the player should be $\frac{\sqrt{A}}{2}$ m away from the width of the pool

12 (iii)

$y=20$ or $A=112$

maximum θ, the player should be $\frac{\sqrt{112}}{2}=2\sqrt{7}$ m away from the width of the pool.

12 (iv)

<u>Using Chain Rule</u>

Since $x=15$, $\tan\theta=\frac{45}{253}$, $A=112$ and $\frac{\mathrm{d}x}{\mathrm{d}t}=-50$,

$$\frac{\mathrm{d}\theta}{\mathrm{d}t}=\frac{\mathrm{d}\theta}{\mathrm{d}x}\times\frac{\mathrm{d}x}{\mathrm{d}t}$$

$$=\frac{12\left(112\right)-48\left(15\right)^2}{\left[4\left(15\right)^2+112\right]^2\left[1+\left(\frac{45}{253}\right)^2\right]}\times-50$$

$$=0.447\text{ rad/min}\left(\text{to 3 sf}\right)$$

13 (i)

Since L lies on the lines of AA' and BB', we need to find the equation of these 2 lines.

Qn	Solution
	$l_{AA'}$: $\mathbf{r} = \begin{pmatrix}2\\3\\3\end{pmatrix} + \lambda\begin{pmatrix}-3-2\\13-3\\8-3\end{pmatrix} = \begin{pmatrix}2\\3\\3\end{pmatrix} + \lambda\begin{pmatrix}-5\\10\\5\end{pmatrix}$ $l_{BB'}$: $\mathbf{r} = \begin{pmatrix}4\\3\\4\end{pmatrix} + \mu\begin{pmatrix}9-4\\13-3\\14-4\end{pmatrix} = \begin{pmatrix}4\\3\\4\end{pmatrix} + \mu\begin{pmatrix}5\\10\\10\end{pmatrix}$ $\begin{pmatrix}2\\3\\3\end{pmatrix} + \lambda\begin{pmatrix}-5\\10\\5\end{pmatrix} = \begin{pmatrix}4\\3\\4\end{pmatrix} + \mu\begin{pmatrix}5\\10\\10\end{pmatrix}$ By G.C., $\lambda = -\frac{1}{5}$ and $\mu = -\frac{1}{5}$ $\overrightarrow{OL} = \begin{pmatrix}2\\3\\3\end{pmatrix} - \frac{1}{5}\begin{pmatrix}-5\\10\\5\end{pmatrix} = \begin{pmatrix}3\\1\\2\end{pmatrix}$
13 (ii)	p_1: $\mathbf{r} \cdot \begin{pmatrix}0\\1\\0\end{pmatrix} = k$ Since $\begin{pmatrix}-3\\13\\8\end{pmatrix} \cdot \begin{pmatrix}0\\1\\0\end{pmatrix} = 13$, $k = 13$. l_{LC}: $\mathbf{r} = \begin{pmatrix}3\\1\\2\end{pmatrix} + \lambda\begin{pmatrix}2-3\\2-1\\5-2\end{pmatrix} = \begin{pmatrix}3\\1\\2\end{pmatrix} + \lambda\begin{pmatrix}-1\\1\\3\end{pmatrix}$ $\left(\begin{pmatrix}3\\1\\2\end{pmatrix} + \lambda\begin{pmatrix}-1\\1\\3\end{pmatrix}\right) \cdot \begin{pmatrix}0\\1\\0\end{pmatrix} = 13$ $1 + \lambda = 13$ $\lambda = 12$ $\overrightarrow{OC'} = \begin{pmatrix}3\\1\\2\end{pmatrix} + 12\begin{pmatrix}-1\\1\\3\end{pmatrix} = \begin{pmatrix}-9\\13\\38\end{pmatrix}$
13 (iii)	Since $(0,1053,0)$ is on p_2, shortest distance is

Qn	Solution
	$\left\lvert \begin{pmatrix} 0-3 \\ 1053-1 \\ 0-2 \end{pmatrix} \bullet \frac{1}{\sqrt{17}} \begin{pmatrix} 0 \\ 1 \\ 4 \end{pmatrix} \right\rvert = \frac{1044}{\sqrt{17}} \approx 253$ cm (3 s.f.)
13 (iv)	Solving $p_1: y=13$ and $p_2: y+4z=1053$, $x=x$ $y=13$. $z=260$ So, l: $\mathbf{r} = \begin{pmatrix} 0 \\ 13 \\ 260 \end{pmatrix} + s\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, where $s \in \mathbb{R}$.